NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR (Computer Science)

FALL SEMESTER EXAMINATIONS 2024

Time : 3 Hours | Batch 2022 | Dated : 09-DEC-24

Max Marks : 60

Ms. Huma

Software Engineering - CT-365

Question Number: 1 *CLO1: Understand* **(12 Marks)**

a. Amna works as a Quality Assurance Engineer at a large electronics company. She is responsible for the final testing of her company's servers and is part of a team which decides when new products will be shipped to distributors for sale. Amna's company has a contract with another company which makes the chips which are incorporated into the servers Amna's company makes. The business model for this product is to release a new generation server approximately every six months, meaning Amna has a limited timeframe to conduct her Quality Control tests. Because there is such a short amount of time between the release of each next new product, the Quality Assurance department cannot perform every possible test on the servers to ensure they are defect free. Amna will not ship a product if there is any possibility that the server could malfunction and cause physical harm to the customer. However, she will ship a product that has a higher likelihood of failure resulting in data loss for the customer, because she knows that if she doesn't, her company's competitor will.

In accordance with the ACM Ethical Principles:

i. ***Express*** if this is an ethical way to conduct business. Which ethical principles are affected? [2]

ii. ***Infer*** how she should determine when to ship a product with known defects. What do the ethical principles state in this regard? [2]

b. You are responsible for the design of a very reliable and fault-tolerant system. ***Discuss*** dependable programming practices that will be suitable for this system. [4]

c. ***Describe*** the major benefits and issues with software reuse. Based on your reasons, express your opinion, either in favor of, or against software reuse. [4]

Question Number: 2 *CLO2: Create* **(12 Marks)**

a. ***Improve*** the following statement of requirements by identifying ambiguities or omissions and rewriting to resolve them. [4]

The home security system will provide comprehensive protection for residences. It will monitor activity inside and outside the house using the advanced technology. The system will notify the homeowner if any suspicious activity is detected. It will also allow remote access to live footage, enabling homeowners to check on their property at any time. Additionally, the system will have a feature to automatically contact emergency services if certain conditions are met. The system will integrate with other smart home devices, offering convenience and additional security features.

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b. *Categorize* the following as functional, non-functional, and domain requirements.

i. In commission-based sales businesses, there is no concept of negative commission. However, if care is not taken novice developers can be lured into developing systems, which calculate negative commission. [1]

ii. A train control system has to take into account the braking characteristics in different weather conditions. [1]

iii. The system should issue an email within 2 seconds of adding new merchandise to the database. [1]

iv. When a site visitor creates an account, the server shall send a welcome email. [1]

c. Adapter pattern is a type of structural design pattern which states that *"Convert the interface of a class into another interface that clients expect. Adapter lets classes work together that could not otherwise because of incompatible interfaces."* *Compose* a code example. [4]

Question Number: 3

CLO3: Apply

(12 Marks)

a. You are testing a gaming application that includes multiplayer functionality. This app should ensure that the players can connect, communicate, and play games together smoothly. ***Produce*** at least four test cases (other than registration and login) for this application. [8]

b. ***Determine*** the benefits of functional test automation. From the test cases generated above, capture at least two test cases which can be automated. [4]

Question Number: 4

CLO3: Apply

(12 Marks)

a. You and several friends are about to prepare a lasagna dinner. The tasks to be performed, their immediate predecessors, and their estimated durations are as follows:

Task	Task Description	Tasks that Must Precede	Time
A	Buy the mozzarella cheese*	—	30 minutes
B	Slice the mozzarella	A	5 minutes
C	Beat 2 eggs	—	2 minutes
D	Mix eggs and ricotta cheese	C	3 minutes
E	Cut up onions and mushrooms		7 minutes
F	Cook the tomato sauce	E	25 minutes
G	Boil large quantity of water		15 minutes
H	Boil the lasagna noodles	G	10 minutes
I	Drain the lasagna noodles	H	2 minutes
J	Assemble all the ingredients	I, F, D, B	10 minutes
K	Preheat the oven		15 minutes
L	Bake the lasagna	J, K	30 minutes

*There is none in the refrigerator.

Construct the project network for preparing this dinner by identifying earliest and latest start and finish times for each activity, slack, and critical path.

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Because of a phone call, you were interrupted for 6 minutes when you should have been cutting the onions and mushrooms. By how much will the dinner be delayed? If you use your food processor, which reduces the cutting time from 7 to 2 minutes, will the dinner still be delayed? [8]

b. ***Explore*** the levels of CMMI in detail. [4]

Question Number: 5

CLO3: Apply **(12 Marks)**

a. ***Solve*** for the estimates of object point count, new object points, and effort to develop a software project that has the following characteristics: [6]

- The application consists of **six screens**, each with **five views** and seven data tables for three servers and four clients.
- It generates **three reports**, each divided into five sections each from seven data tables for two servers and three clients.
- The application will comprise of two **3GL** components.
- Approximately **20%** of the **screens and reports** are reused from a previous project.
- Developer's experience and capability in similar environment is nominal.

b. A new extremely reliable project for monitoring heart rate, with estimated 400 KLOC has to be developed for continuously measuring a patient's heart rate. The team has the developers with a lot of programming language experience. The project manager has to estimate the duration of this project, but is confused with the decision on whether to incorporate modern programming practices or not. ***Illustrate*** your answer by calculating duration for both cases. [6]

TABLES FOR REFERENCE ARE PROVIDED AT PAGE 4.

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Tables for Question Number: 5 (a)

No. of views contain	Sources of data tables		
	Total < 4 (< 2 servers < 3 clients)	Total < 8 (2 - 3 servers 3-5 clients)	Total 8 + (> 3 servers > 5 clients)
< 3	Simple	Simple	Medium
3 - 7	Simple	Medium	Difficult
> 8	Medium	Difficult	Difficult

For Screens

No. of section contain	Sources of data tables		
	Total < 4 (< 2 servers < 3 clients)	Total < 8 (2 - 3 servers 3-5 clients)	Total 8 + (> 3 servers > 5 clients)
0 - 1	Simple	Simple	Medium
2 - 3	Simple	Medium	Difficult
4 +	Medium	Difficult	Difficult

For Reports

Object Type	Complexity Weight		
	Simple	Medium	Difficult
Screen	1	2	3
Report	2	5	8
3GL Components	-	-	10

Complexity Weight

Developers experience & capability	Productivity (PROD)
Very Low	4
Low	7
Nominal	13
High	25
High	50

Productivity Rate

Tables for Question Number: 5 (b)

Project	a_i	b_i	c_i	d_i
Organic	3.2	1.05	2.5	0.38
Semi-detached	3.0	1.12	2.5	0.35
Embedded	2.8	1.20	2.5	0.32

Cost Drivers	RATINGS					
	Very low	Low	Nominal	High	Very High	Extra High
Product Attributes						
RELY	0.75	0.88	1.00	1.15	1.40	-
DATA	-	0.94	1.00	1.08	1.16	-
CPLX	0.70	0.85	1.00	1.15	1.30	1.65
Computer Attributes						
TIME	-	-	1.00	1.11	1.30	1.66
STOR	-	-	1.00	1.06	1.21	1.56
VIRT	-	0.87	1.00	1.15	1.30	-
TURN	-	0.87	1.00	1.07	1.15	-

Cost Drivers	RATINGS					
	Very low	Low	Nominal	High	Very high	Extra high
Personnel Attributes						
ACAP	1.46	1.19	1.00	0.86	0.71	-
AEXP	1.29	1.13	1.00	0.91	0.82	-
PCAP	1.42	1.17	1.00	0.86	0.70	-
VEXP	1.21	1.10	1.00	0.90	-	-
LEXP	1.14	1.07	1.00	0.95	-	-
Project Attributes						
MODP	1.24	1.10	1.00	0.91	0.82	-
TOOL	1.24	1.10	1.00	0.91	0.83	-
SCED	1.23	1.08	1.00	1.04	1.10	-

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SEAT NO. CT-21090

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(Bachelor of Science in Computer Science & Information Technology)

FALL SEMESTER EXAMINATIONS 2023

Time : 3 Hours

Batch 2021

Dr Shariq Mahmood

Dated : 26-JAN-24

Max Marks : 60

Software Engineering - CT-365

Instructions: Attempt all questions. Each question carries equal marks.

Q.1 Consider a software project that requires estimation using different models: Lines of Code (LOC), Function Points (FP), and COCOMO model. The project specifications indicate a complexity that translates to 40,000 lines of code with the average labor cost is Rs.100,000/- per month. (10) CLO-5

[i] Calculate the total estimated project cost and estimated effort required for the project using the LOC model, assuming an average of 600 lines of code per person-month.

[ii] Calculate the function point, total estimated project cost and estimated effort required software application using FP model with multiple Processing Factors 5, 1, 0, 4, 3, 5, 4, 3, 4, 5, 2, 3, 4, 2 by using following given Date:

Information Domain	Weight	Estimated Value
The number of EI	4	22
The number of EO	4	45
The number of EQ	6	06
The number of ILF	10	05
The number of EIF	5	02

Average Productivity for this type of Application is 6.5 FP / Per Month

[iii] Calculate the total estimated project cost and estimated effort required for the project using COCOMO model. The project attributes to be in the 'organic' mode with an A=2.4, B=1.05, and C=2.5.

Relate and discuss the results obtained from the three estimation models in terms of estimated cost and the effort required.

Q.2 Discuss the challenges associated with choosing an appropriate SDLC model for a software project. Explain the phases of the Spiral model in SDLC. **Illustrate** with an example how the Spiral model differs from other traditional SDLC models and its significance in managing complex projects. (10) CLO-3

Q.3 **Explain** the concept of Layered Technology in Software Engineering, outlining the various layers involved and their significance in the software development process. Additionally, describe the umbrella activities and principles of Software Engineering. (10) CLO-2

Q.4 Design a comprehensive use case diagram for a University Portal, integrating the process of requirement gathering into the design: (10) CLO-5

[i] Propose structured approach to gather requirements for the University Portal system. With your own observations identify the system functionalities and consider these identified functionalities as gathered requirements.

[ii] Create a use case diagram that reflects the outcomes of the requirement gathering process. Demonstrate at least five primary actors and their relationships within the system, emphasizing how these were identified through requirement elicitation methods.

[iii] Present a minimum of eight distinct use cases derived from the gathered requirements, ensuring each use case is well-defined, linked to associated actors, and aligns with the stakeholders' needs and system functionalities.

[iv] Evaluate the significance of requirement gathering in the context of use case diagram development, emphasizing how it ensures accurate representation of user needs and facilitates effective communication among project stakeholders during system design and development.

Q.5 **Illustrate** the following questions with reasoning (10) CLO-3

[i] Using your own words, describe the difference between verification and validation with the help of real life example.

[ii] Discuss Software Testing Strategies in detail. Is unit testing possible or even desirable in all circumstances? Provide examples to justify your answer

[iii] List some problems that might be associated with the creation of an independent test Group (ITG). Differentiate between the strategy of testing the software by the developer and the tester.

Q.6 **Explain** the following with example. (10) CLO-2

[i] Agile Software Development [iii] Scrum Process

[ii] Software Quality Assurance [iv] 4 P's of Project Management

SEAT NO. CT-30032.

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(BACHELOR OF SCIENCE IN COMPUTER SCIENCE)
FALL SEMESTER EXAMINATIONS 2022
BATCH 2021

Time: 3 Hours | **Dated:07-02-2023**
Max.Marks:60

Software Engineering - CT-365

Note: Attempt all questions. Refer to Scenarios 1 & 2 given at the end of the paper, if needed in any question.

Q. No 1	*CLO-1 (C2)*	
(a)	***Give an example*** of a form-based structured specification of the Polio Vaccine Software System (Scenario 1). This should include different fields such as *Function, Description, Input source, Outputs, Destination, Action, Requires, Precondition, Postcondition, and Side effect.*	4
(b)	***Describe*** the challenges related to the *Software Requirements Elicitation* in the case of Polio Vaccine Software System (Scenario 1).	4
(c)	***Explain*** the use of automated test components. Also give examples of scenarios when these components should be used.	4
Q. No 2	*CLO-3 (C3)*	
(a)	***Apply*** the idea of *Test-First development* to highlight possible problems that may arise as a result of this type of development.	4
(b)	***Demonstrate*** how the *context model* helps in defining the *external perspective* of a Polio Vaccine Software System (Scenario 1).	4
(c)	In order to describe the *interaction modeling* of the Polio Vaccine Software System (Scenario 1) ***construct*** a use-case diagram and a *sequence* diagram of your project.	4
Q. No 3	*CLO-3 (C3)*	
(a)	To reduce costs and the environmental impact of commuting, your company decides to close a number of offices and to provide support for staff to work from home. However, the senior management who introduced the policy is unaware that software is developed using Scrum. ***Employ*** the use of technology to support Scrum in a distributed environment to make this possible. Explore the problems that you are likely to encounter using this approach.	4
(b)	***Apply*** COCOMO cost modeling to obtain estimated cost of the Polio Vaccine Software System (Scenario 1).	4
(c)	As a project manager responsible for the development of the Polio Vaccine Software System (Scenario 1), ***capture*** the resources you would require to complete this project.	4
Q. No 4	*CLO-2 (C6)*	
(a)	***Design*** the architecture of Polio Vaccine Software System (Scenario 1).	4
(b)	***Formulate*** the non-functional requirements of the Polio Vaccine Software System (Scenario 1) by defining a few *properties* with their *metrics* and *measures*.	4
(c)	With the help of examples describe the differences between software *verification* and *validation*. Also, ***generate*** a model of the software testing process with the help of a diagram.	4

Q. No 5	CLO-2 (C6)
(a)	***Compile*** an analysis of the given security story (Scenario 2) regarding the *Finance Software System* with respect to each of the following security terminologies: Asset, Attack, Control, Exposure, Threat, Vulnerability.
(b)	***Compose*** a thorough review about the challenges in the cost estimation of the Polio Vaccine Software System (Scenario 1) based on your perspective.
(c)	***Develop*** risk assessment report of the Polio Vaccine Software System (Scenario 1) in tables of at least three entries: (i) An asset analysis with assets in the system and their value. (ii) Threat and control analysis that a system may face.
	Scenario 1: Polio Vaccine Software System
	The Government aims to launch a mobile application in all cities to identify and vaccinate children possibly missed in polio immunization drives. The application will be given to the polio eradication teams who will upload a picture of a child in the application along with the house number and send it to the dashboard connected with a smartphone (some people may have an objection to taking a picture). Once uploaded, household data of children in various campaigns is available on the dashboard and cross-matched for verification. Any ambiguity in the number of children from previous campaigns is investigated in the field. The mobile application will have several features including a 'refusals' column, household number, and number of children. The data recorded through the mobile app will be monitored at multiple layers by the supervisors at the highest level and if any discrepancy will be found an investigation will be launched. Any citizen may also install this mobile app to find, track and report any correction of the data related to his family.
	Scenario 2: Finance Software System
	Finance department staff log on to the Finance Software System using the username and a password. The system requires passwords to be at least eight letters long but allows any password to be set without further checking. There is no check on the number of failed password attempts to lock the account. A criminal finds out that a well-paid employee is receiving salary with a handsome amount of bonus. He would like to gain illegal access to information in this system so that he can blackmail the employee. By posing as a concerned relative and talking with the staff member of Finance department, he discovers how to access the system and personal information about the employees and their families. By checking name badges, he discovers the names of some of the people allowed access. He then attempts to log on to the system by using these names and systematically guessing possible passwords, such as the names of the employee's children etc.

** THE END **

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY
DEPARTMENT OF COMPUTER SCIENCE & INFORMATION TECHNOLOGY
TSCS, Midterm Examinations Fall 2022

Time: 90 minutes **Software Engineering (CT-365) - B** **Max Marks: 20**
Date: 05-12-2022

Note: Attempt all questions.

Q-1) a) ***Explain*** the ethical responsibilities of a professional Software Engineer with appropriate real life ***examples*** and scenarios. [CLO 1] [3]

b) ***Indicate*** the deliverables of: 1. Planning Phase 2. Module Testing and implementation. [CLO 1] [1]

c) ***Differentiate*** between *Requirements Specification* and *Requirements Refinement* stages in the Reuse oriented Software Engineering. [CLO 1] [2]

Q-2) ***Discuss*** the scenarios for which incremental development is favorable. ***Illustrate*** its pros and cons. [CLO 1] [5]

Q-3) A group diary and online system is intended to support the scheduling of meetings and appointments across a group of coworkers. When an appointment is to be made that involves a number of people, the system finds a common slot in each of their diaries and arranges the appointment for that time. If no common slots are available, it interacts with the user to rearrange the appointment in his or her personal diary to make time for the appointment. ***Compose*** the *functional and non-functional requirements* for the system. [CLO 2] [5]

Q-4) ***Design*** a *Sequence* diagram and an *Architecture* diagram for the above-mentioned scenario. [CLO 2] [5]

SEAT NO. _____

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

THIRD YEAR(BACHELOR OF SCIENCE IN
(COMPUTER SCIENCE & INFORMATION TECHNOLOGY)
SPRING SEMESTER EXAMINATIONS 2021
BATCH 2018

Dated:25-08-2021

Time: 3 Hours

Max.Marks:60

Software Engineering - CT-365

Note: Attempt all questions. All questions/parts carry equal marks.

Q. No 1

(a) *Clinic staff log on to the Mentcare system using a username and password. The system requires passwords to be at least eight letters long but allows any password to be set without further checking. A criminal finds out that a well-paid sports star is receiving treatment for mental health problems. He would like to gain illegal access to information in this system so that he can blackmail the star. By posing as a concerned relative and talking with the nurses in the mental health clinic, he discovers how to access the system and personal information about the nurses and their families. By checking name badges, he discovers the names of some of the people allowed access. He then attempts to log on to the system by using these names and systematically guessing possible passwords, such as the names of the nurses' children.* 4

For the above given security story regarding the *Mentcare System* explain each of the security terminologies: Asset, Attack, Control, Exposure, Threat, Vulnerability.

(b) Describe with diagram, the risk-driven security requirements process with different stages from *Asset Identification* to *Security Requirement Definition*. 4

(c) For a preliminary risk assessment report of Insulin Pump System describe in tables of at least three entries: (i) An asset analysis with assets in the system and their value. (ii) Threat and control analysis that a system may face. 4

Q. No 2

(a) Explain the idea of *Test-First development* and highlight possible problems that may arise as a result of this type of development. 4

(b) Explain with an example, how the *context model* helps in defining the *external perspective* of a software system that you committed to develop in the classroom activity. 4

(c) In order to describe the *interaction modeling* of the software project that you committed to 4

Question#3 **(12 marks)**

DRAW a Use case diagram/Sequence diagram for the following situation (state any assumptions you believe you have to make in order to develop a complete diagram). Then convert the use case diagram into a sequence diagram. The H. I. Topi School of Business operates international business programs in ten locations throughout Europe. The school had its first class of 9000 graduates in 1965. The school keeps track of each graduate's student number, name, country of birth, current country of citizenship, current name, current address, and the name of each major the student completed (each student has one or two majors). To maintain strong ties to its alumni, the school holds various events around the world. Events have a title, date, location, and type (e.g., reception, dinner, or seminar). The school needs to keep track of which graduates have attended which events. When a graduate attends an event, a comment is recorded about the information school officials learned from that graduate at that event. The school also keeps in contact with graduates by mail, e-mail, telephone, and fax interactions. As with events, the school records information learned from the graduate from each of these contacts. When a school official knows that he or she will be meeting or talking to a graduate, a report is produced showing the latest information about that graduate and the information learned during the past two years from that graduate from all contacts and events the graduate attended.

Question#4 **(12 marks)**

Imagine an information system built without using a systems analysis and design methodology and without any thinking about the SDLC. Use your imagination and d escribe any and all problems that might occur, even if they seem a bit extreme or illogical. (The problems you will describe have probably occurred in one setting or another.). SUGGEST new requirements to create/update that information system. WHAT will be your initial steps in creation/modification? WHICH process model will suit this case if there is no time-dependency and WHY?

Question#5 **(12 marks)**

A project consists of activities from A-H as shown in the following table. The immediate predecessor(s) and the duration in weeks of each of the activities are given in the same table. DRAW the project network and FIND the critical path and the corresponding project completion time. Also FIND the total float/slack as well as free float for each of the non-critical activities.

Activities	Description	Duration (weeks)	Predecessor(s)
A	Collect Requirements	7	-
B	Analyze data	9	-
C	Design Data	12	A
D	Design Screens & Reports	8	A, B
E	Program	9	D
F	Test and Document	6	C, E
G	Install	5	E

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR FALL SEMESTER (SOFTWARE ENGINEERING)
EXAMINATIONS 2017-18
BATCH 2016-2017

Time: 3 Hours | **Dated:16-03-2018**
Max.Marks:60

Software Engineering - SE-207

Instructions: Answer 5 questions
All questions carry equal marks

Q#1.

a. You have been appointed as a project manager within an information systems organization. Your job is to build an application that is quite similar to others your team has built, although this one is larger and more complex. Requirements have been thoroughly documented by the customer. What software process model(s) you choose and why? [03]

b. List all the designs concepts and give description for each [09]

Q#2.

a. Discuss how structural partitioning can help to make software more maintainable. [03]

b. Write the advantages and disadvantages of waterfall and prototyping model. [06]

c. What are the elements of analysis model? [03]

Q#3.

a. Assume that you are the project manager for a company that builds software for consumer products. You have been contracted to build the software for home security system. Write a statement of scope that describes the software. [04]

b. Define software requirement engineering and describe the activities of software requirement engineering. [05]

c. Explain the inputs and outputs of Domain analysis [03]

Q#4.

a. Develop a Use case diagram, Use case text and Data flow Diagram for network based course registration system of university. [08]

b. Discuss software engineering as a layered technology. [04]

Q#5.

a. Propose a specific software project that would be amenable to the spiral model, present a scenario for applying the model to the software. [04]

b. Differentiate [06]

i. Black box and White box testing

ii. Functional and Non-functional requirements

iii. Reactive and Proactive risk strategies

c. List the Lean software Development principles [02]

Q#6.

a. Define agile methodology also discuss scrum in detail [05]

b. Is unit testing possible or even desirable in all circumstances? Provide examples to justify your answer [02]

c. Discuss different testing strategies used in software engineering. [05]

— X —

NED UNIVERSITY OF ENGINEERING & TECHNOLOGY

SECOND YEAR SPRING SEMESTER (SOFTWARE ENGINEERING)
EXAMINATIONS 2018
BATCH 2016-2017

Time: 3 Hours | **Dated:28-08-2018**
Max.Marks:60

Software Requirement Engineering - SE-208

40+33
73

(1) Attempt all questions.
(2) All questions carry equal marks.
(3) Give examples and draw diagrams where necessary

Q.No.1

4+4.5

(a) Define goals? Explain the advantages of using goal-oriented requirements engineering? **(8+6+6)**

(b) Define functional and non-functional goals. Define different goal categories **(CLO3)**

(c) Construct a goal graph of a Library Management System. The parent goal from where you should start is ***Achieve[BookRequestSatisfied]*** **(CLO3)**

Q.No.2

(a) Define stakeholder-driven requirements elicitation and artefact-driven requirements elicitation methods. Explain in detail different methods of artifact-driven requirements elicitation. **(10)** 6 **(CLO2)**

Q.No.3

3+2

(a) Define **(4+6)**
i. Fault trees **(CLO1)**
ii. Product-related risks

(b) Construct a security threat tree for the risk of denial of service incurred by your favourite e-mail server. **(CLO1)**

Q.No.4

5+3

(a) Explain the classical requirements engineering process. **(5+5)** 7 **(CLO2)**

(b) Discuss a standardized template for organizing a requirements document. **(CLO2)**

Q.No.5

8

(a) Define Obstacles. Discuss why we liken obstacles to risks. Consider the goal *AuthenticatedAccess* is obstructed by the obstacle *PasswordForgotten* in a Web service. Justify that this obstacle is the indication of a risk and has to be mitigated. Explore goal restoration countermeasures to this obstacle. **(10)** **(CLO1)**

——— X ———